Clustering with K-means

ES 335 / ES 678 Machine Learning, Sem I 2026 — Prof. Udit Bhatia Sources: `Lectures/L13_kmeans_annotated_slides.pdf` (35 slides), `Lectures/L13a_kmeans_handwritten_class_notes.pdf` (3 pages), `Lectures/L13b_unsupervised_learning_handbook.pdf` (Chapter 1 and Chapter 2)

What this file is really about:

$$\text{no labels} \rightarrow \text{define "together"} \rightarrow \text{minimise } J \rightarrow \text{start well} \rightarrow \text{choose } K \rightarrow \text{know when it fails}$$

How to use this file. Everything runs on one small dataset, eight training runs, and every number can be checked by hand. Part A builds the mathematics from the start. Part B walks through all 35 slides and the handwritten notes. Part C is practice with answers.

The one dataset for the whole lecture (x_1 is training time in hours, x_2 is peak GPU memory in GB):

run	1	2	3	4	5	6	7	8
x_1	4	4	2	1	8	8	8	6
x_2	4	3	3	3	8	6	9	8

Step-up path

You need. Distance between two points, the mean, and the idea of a derivative.

1. Say what "together" means when there are no labels.
2. Write the objective, and derive why each centroid is a mean and each boundary a straight line.
3. Run Lloyd's algorithm by hand, and see why it stops but may stop badly.
4. Start well with K-means++, and standardise the features.
5. Choose K with several scores, and name where K-means fails.

You can stop when you can run K-means by hand, prove that J never rises, and name data that break it.

Contents

Part A Maths From the Ground Up

A1 Learning without labels

In **supervised** learning each example comes with an answer: we are given pairs (x_i, y_i) and learn a function $f(x) \to y$. In **unsupervised** learning we are given only the x_i, and we ask which examples belong together.

There is no y, so there is no error to measure against a correct answer. A child sorting LEGO might sort by colour, size or shape, and each is a reasonable answer. The method has to be told what "together" means, and that choice is the whole model.

An example is a list of numbers called **features**, so it is a point in a space with one axis per feature. A **cluster** is a group of points that lie close together in that space.

A2 Distance between two points

The straight-line or **Euclidean distance** between x and y comes from Pythagoras, applied one coordinate at a time:

$$\|x - y\| = \sqrt{\sum_j (x_j - y_j)^2}.$$

> **Example.** Runs 1 and 8 are $(4, 4)$ and $(6, 8)$. The differences are 2 and 4, so the squared distance is $4 + 16 = 20$ and the distance is $\sqrt{20} \approx 4.47$.

Squared distance adds one term per feature. So a feature measured on a large scale contributes large terms and dominates, and that fact returns in A12.

A3 Mean, variance and standardisation

The **mean** of m numbers is their sum over m. Their **variance** is the average squared distance from the mean, and the **standard deviation** is its square root.

To **standardise** a feature, subtract its mean and divide by its standard deviation. The result has mean 0 and variance 1, and has no units.

A4 The best single number: why the mean

Suppose one cluster has values $x_1, \dots, x_m$ in one coordinate, and we must pick one number c to represent them. Measure the cost by squared distance:

$$f(c) = \sum_{j=1}^{m} (x_j - c)^2.$$

Step 1 — Differentiate. $f'(c) = -2 \sum_j (x_j - c)$.

Step 2 — Set to zero. $\sum_j x_j - mc = 0$, so $c = \frac{1}{m} \sum_j x_j$, the mean.

Step 3 — Check it is a minimum. $f''(c) = 2m > 0$.

> **Example.** The small runs have first coordinates $4, 4, 2, 1$, with mean 2.75. Then $f(2) = 9$, $f(2.75) = 6.75$ and $f(3.5) = 9$.

Using absolute distance instead gives the **median**, and that version is called K-medians. The mean is cheap to compute but pulled by outliers: $1, 2, 3, 100$ has mean 26.5 and median 2.5.

A5 The K-means objective

Assign each example a label $g_i \in \{1, \dots, K\}$ and give each cluster a representative point μ_k, its **centroid**. Minimise the total squared distance from each point to its own centroid:

$$J = \sum_{i=1}^{n} \|x_i - \mu_{g_i}\|^2.$$

There are two sets of unknowns, the labels and the centroids. Trying every labelling is hopeless, since there are K^n of them: 2^{1000} has 302 digits.

> **Intuition.** A pizza company opens K outlets. Households are points, outlets are centroids, and J is the total squared delivery distance.

partition of the eight runs	clusters	J
all eight together	1	100.875
a mixed split $\{1,2,5,8\},\{3,4,6,7\}$	2	99.25
small and large $\{1,2,3,4\},\{5,6,7,8\}$	2	15.25

The key observation: **each unknown is easy when the other is fixed.**

A6 Lloyd's algorithm

Repeat until no label changes:

1. **Assign.** Each point joins its nearest centroid: $g_i \leftarrow \arg\min_k \|x_i - \mu_k\|^2$.
2. **Update.** Each centroid moves to the mean of its points.

An assignment step costs about nKd operations, and the number of iterations is usually small, so it works on millions of points. If a cluster becomes empty its mean is undefined: implementations either leave its centroid alone or move it to the point farthest from its own centroid.

A7 A complete worked run

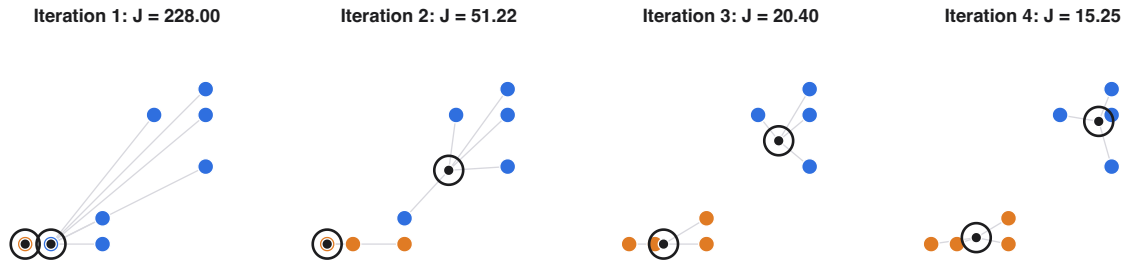
Start deliberately badly, with $\mu_A = (2,3)$ and $\mu_B = (1,3)$, both among the small runs.

Iteration 1, assign. Every run but run 4 is nearer to A, so $J = 228.00$.

Iteration 1, update. A holds runs 1, 2, 3, 5, 6, 7, 8, with mean $(40/7,\ 41/7) = (5.714,\ 5.857)$. B holds run 4 alone, so it stays at $(1,3)$.

Iteration 2, assign. Runs 2 and 3 are now closer to B. $J = 51.22$.

Iterations 3 and 4. Run 1 moves to B too, $J = 20.40$, and then $J = 15.25$. Nothing changes after that.



Lloyd's algorithm on the eight runs from a poor start. Colours are the assignment in each iteration and rings are the centroids used for it. The objective printed above each panel falls at every step, and the final partition is the best of all 256.

A8 The boundary between two clusters

A point x is nearer to μ_a than to μ_b when $\|x-\mu_a\|^2 < \|x-\mu_b\|^2$. Expand both sides. The term $\|x\|^2$ cancels and what remains is **linear** in x:

$$2(\mu_b-\mu_a)^\top x < \|\mu_b\|^2 - \|\mu_a\|^2.$$

The boundary is a straight line, in fact the **perpendicular bisector** of the two centroids. For the final centroids of the eight runs it is $9.5x_1 + 9x_2 = 98.1875$. With K centroids each region is convex, which is why K-means can only ever produce straight boundaries.

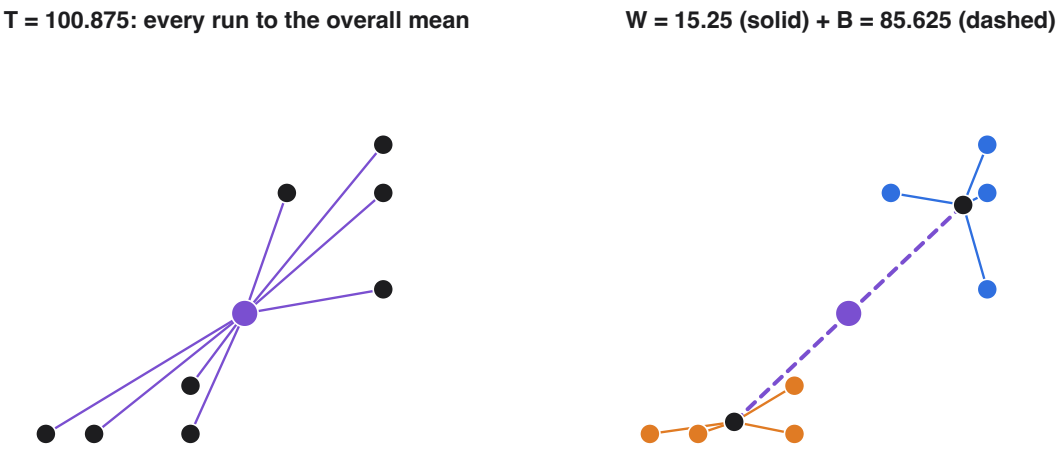
A9 Within-cluster and between-cluster spread

For any partition, the total spread splits exactly in two:

$$\underbrace{\sum_i \|x_i-\bar{x}\|^2}_{T} = \underbrace{\sum_i \|x_i-\mu_{g_i}\|^2}_{W=J} + \underbrace{\sum_k n_k\|\mu_k-\bar{x}\|^2}_{B}.$$

Why. Add and subtract μ_{g_i} inside the first norm. The cross term sums to zero within each cluster, since the deviations from a cluster mean sum to zero.

T is fixed by the data, so **a smaller W is the same thing as a larger B**. For the eight runs, $100.875 = 15.25 + 85.625$. The factor n_k is there because each cluster mean stands for n_k points.

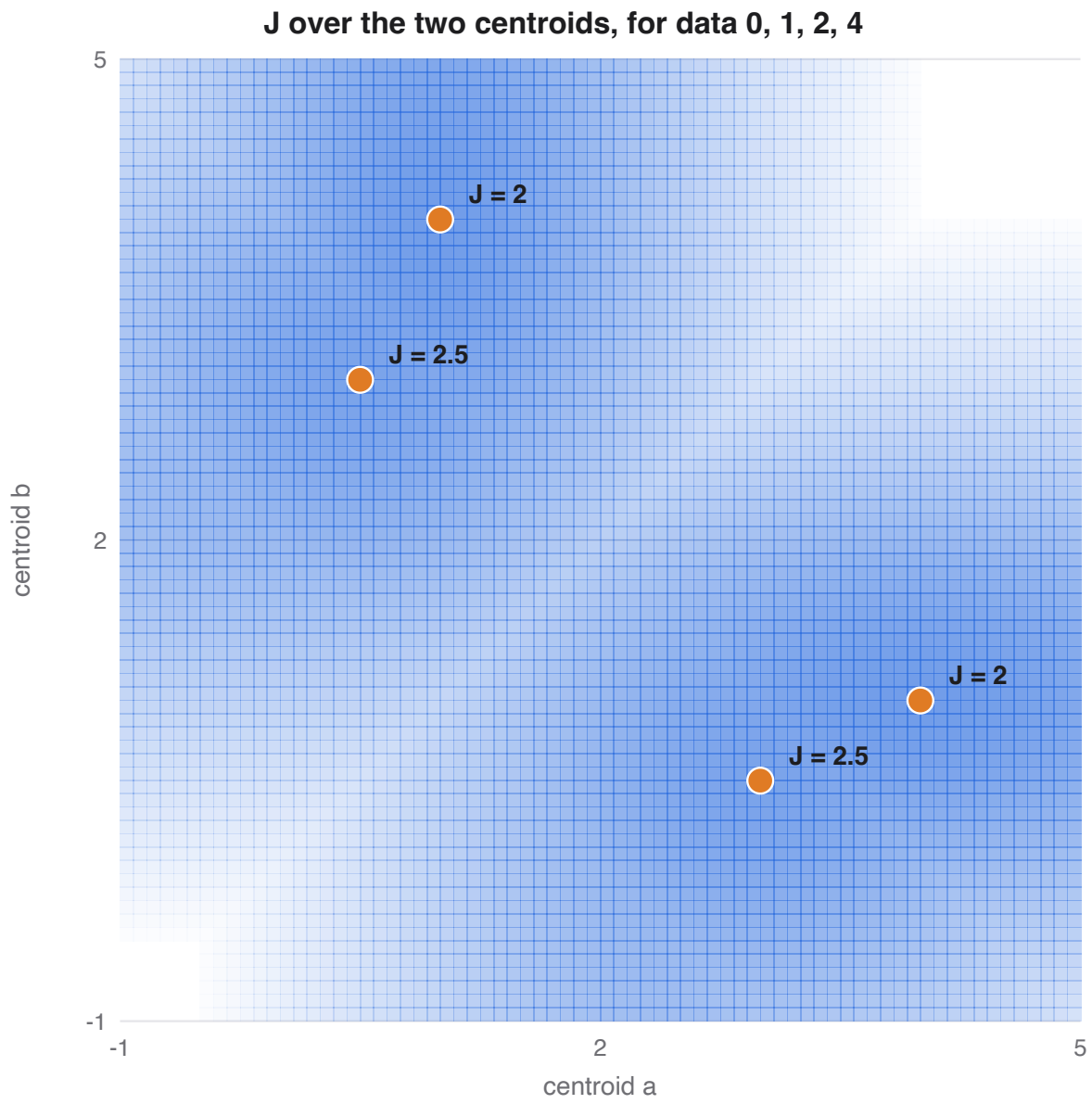


The decomposition $T = W + B$ on the eight runs. Left: every run to the overall mean. Right: every run to its cluster mean (solid, W), and each cluster mean to the overall mean (dashed, counted once per point, B).

A10 Why it stops, and why that is not enough

Neither step can raise J. The assignment gives each point its closest centroid, and the update uses the mean, which is the best centroid for those points. There are finitely many labellings and J can never return to an old value without being unchanged, so the algorithm stops.

> **Careful.** Stopping does not mean best. For the numbers $0, 1, 2, 4$ with $K = 2$, Lloyd can stop at $\{0, 1\}, \{2, 4\}$ with $J = 0.25 + 0.25 + 1 + 1 = 2.5$, though $\{0, 1, 2\}, \{4\}$ has $J = 1 + 0 + 1 + 0 = 2$. Both are **local minima**: no step improves either. Finding the global minimum is NP-hard.



The objective as a function of the two centroids for the data $0, 1, 2, 4$, where darker is lower. There are separate valleys, so where you start decides where you stop.

A11 Starting well: K-means++

With 90 points in three equal groups, the three random starting points land in three different groups with probability $\frac{60}{89} \times \frac{30}{88} = 0.230$. So 77% of random starts put two centroids in one group, which can end at $J = 453.1$ instead of the best 77.6.

K-means++ spreads the starts:

1. Choose the first centroid uniformly from the data.
2. For each point let D_i be the squared distance to its nearest chosen centroid.
3. Choose the next centroid to be point i with probability $D_i / \sum_l D_l$. Repeat.

Why not simply take the farthest point? Because the farthest point is usually an outlier, and a centroid would be wasted on it. A random draw gives one outlier only its own share, while thirty moderately far points together outweigh it. Over 400 starts on the 90 points, the best partition was reached 84% of the time from random starts and 97.8% from K-means++.

> **Example.** For data $0, 1, 10$ with first centroid 0, the values D are $0, 1, 100$. So 1 is chosen with probability $1/101$, and 10 with probability $100/101$.

A12 Standardise the features

Squared distance adds across features, so units decide what matters. On 80 runs recorded in hours and megabytes, K-means split on memory and agreed with the true groups only 51% of the time. After standardising, it agreed 100%.

Standardising is a modelling choice: it says every feature should have equal influence. If some are known to be more important, select or weight them instead.

A13 How many clusters?

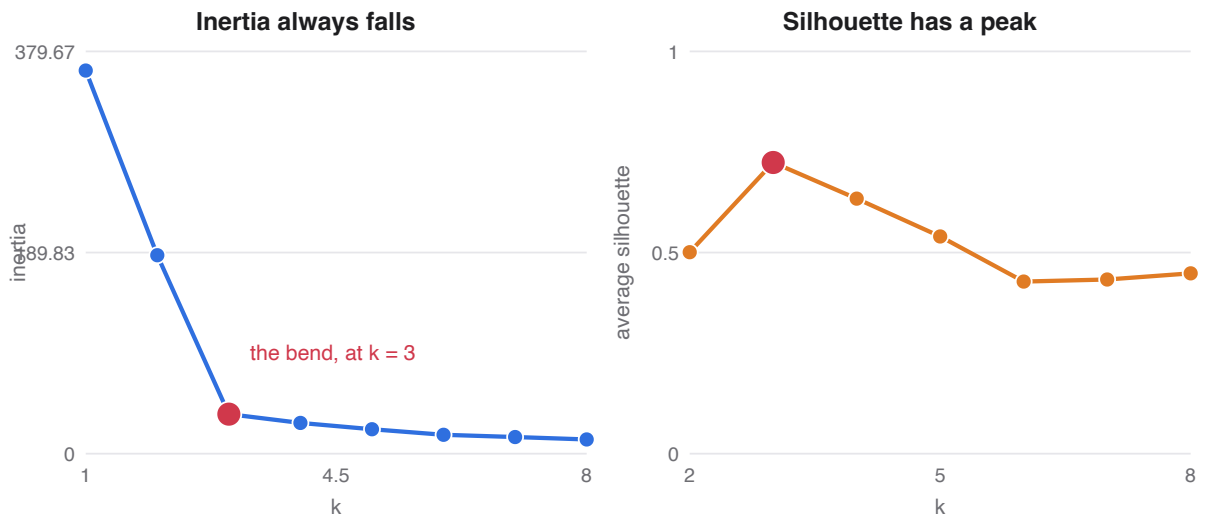
K must be chosen before running, and J cannot choose it, because more clusters always gives a smaller or equal best J, reaching 0 at $K = n$.

K	1	2	3	4	5	6	7	8
best J for the eight runs	100.875	15.250	8.750	4.333	1.500	1.000	0.500	0.000

Five approaches:

- **Elbow.** Plot the best J against K, and find where the fall slows. Often there is no clear bend.
- **Silhouette.** For point i, let a_i be its mean distance to the rest of its own cluster and b_i the mean distance to the nearest other cluster. Then $s_i = (b_i - a_i)/\max(a_i, b_i)$. Values near 1 mean well placed, near 0 on a border, and below 0 probably wrong. Pick the K with the largest mean.
- **Calinski-Harabasz.** Between-spread per degree of freedom over within-spread per degree of freedom: $\mathrm{CH}(K) = \dfrac{B/(K-1)}{W/(n-K)}$. Pick the largest. The divisors matter because B grows and W shrinks with K regardless.

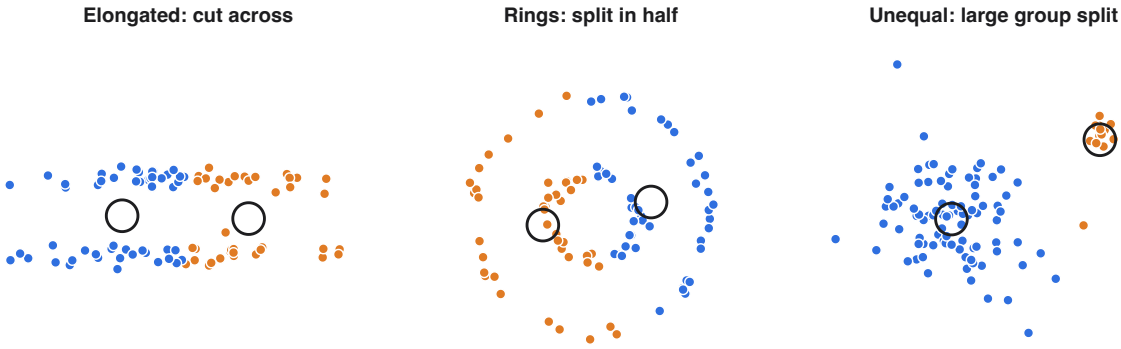
- **Gap statistic.** $\mathrm{Gap}(K) = \overline{\log W_K^*} - \log W_K$, where W_K^* comes from uniform random data in the same box. Pick where the data beat random data by the most.
- **Domain knowledge.** Three marketing budgets means $K = 3$. Often the application decides.



Two ways to read the same runs. Inertia always falls, so look for the bend. The silhouette can go down as well as up, and here it peaks at the true number of groups.

A14 Where K-means fails

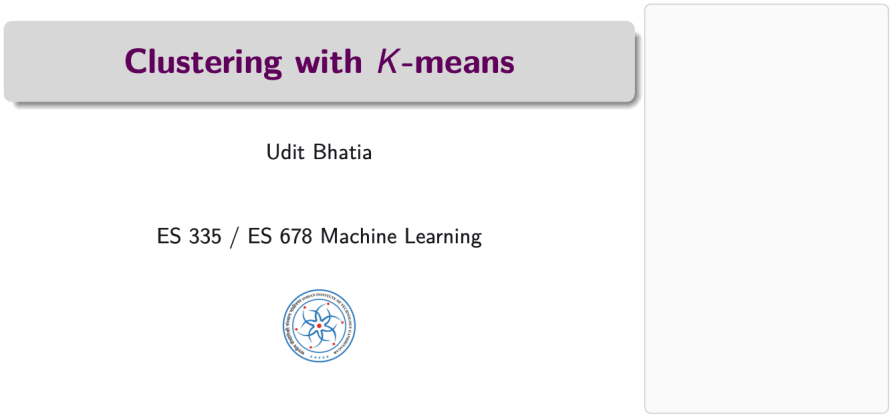
Limitation	Where it bites
K must be chosen	data with no clear groups
The result depends on the start	a random start splits a group
It depends on feature units	hours against megabytes
Outliers pull centroids	one extreme run drags a mean
Round, similar clusters and straight boundaries	long clusters, half-moons
Numbers and Euclidean distance only	categories such as GPU type



Colours show what K-means finds with two clusters. Each shape has two true groups, and each is cut wrongly because K-means prefers round groups of similar size.

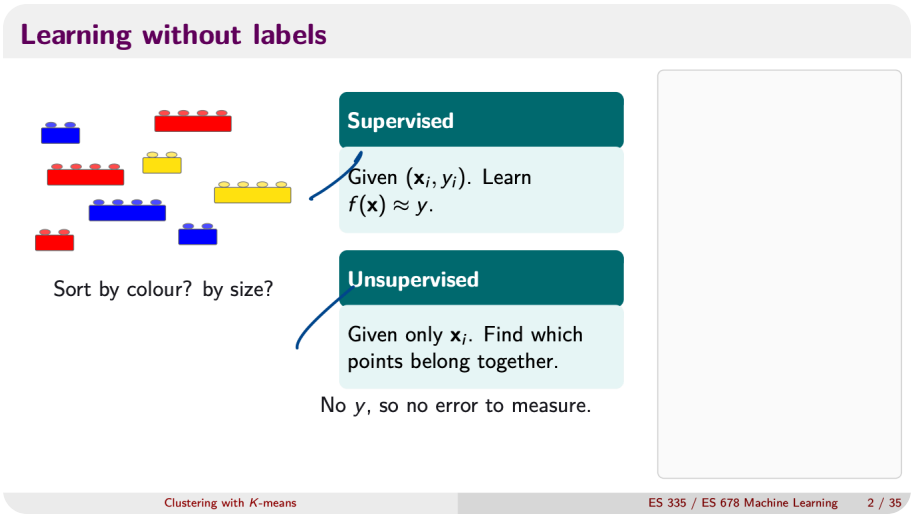
Part B Slide by Slide

Slide 1 Title



Slide 1: the lecture title.

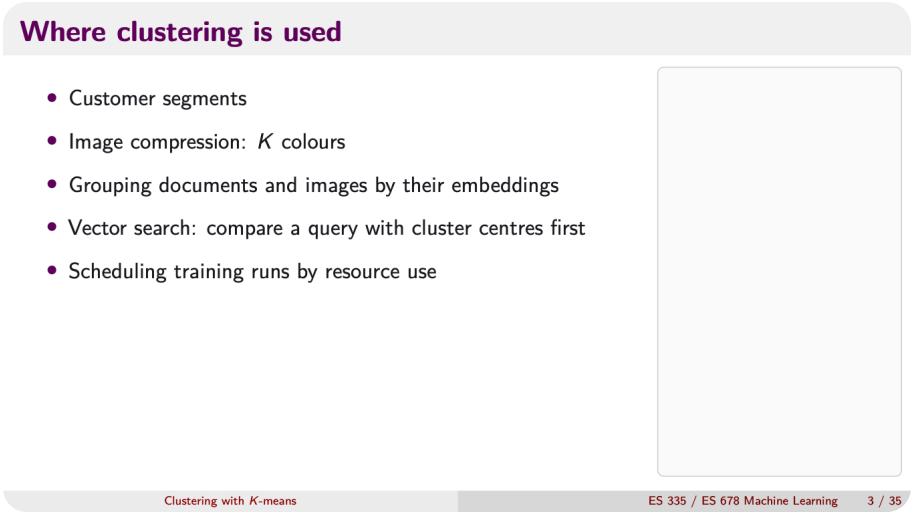
Slide 2 Learning without labels



Slide 2: the same objects could be sorted by colour or by size. Supervised learning has a y to measure against, and unsupervised learning has no y.

Unsupervised learning finds which points belong together. See A1.

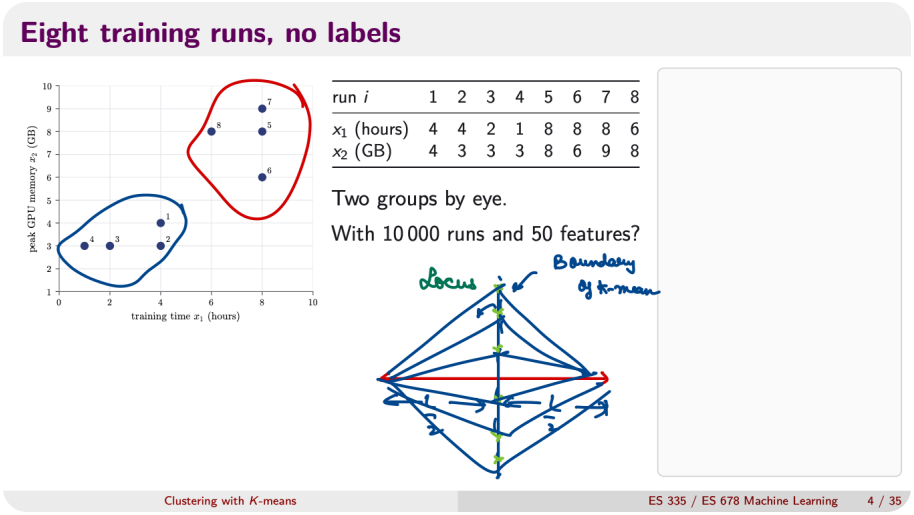
Slide 3 Where clustering is used



Slide 3: customer segments, image compression with K colours, grouping embeddings, vector search and scheduling training runs by resource use.

Vector search compares a query with cluster centres first, so it only searches inside the closest clusters.

Slide 4 Eight training runs, no labels



Slide 4: the running dataset. Two groups are obvious by eye here, but not with 10,000 runs and 50 features.

Slide 5 The K-means objective

The K-means objective

Choose labels $g_i \in \{1, \ldots, K\}$ and centroids $\boldsymbol{\mu}_k$ to minimise

$$J = \sum_{i=1}^{n} \|\mathbf{x}_i - \boldsymbol{\mu}_{g_i}\|^2$$

$\frac{\partial J}{\partial \mu} = 0$

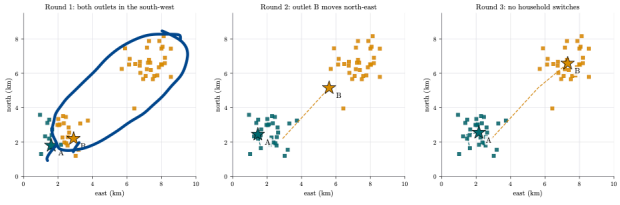
- Two unknowns: labels and centroids
- All labellings: K^n (2^{1000} has 302 digits)
- Fix one unknown $\Rightarrow$ the other is easy

Slide 5: two unknowns, labels and centroids, and K^n possible labellings.

See A5.

Slide 6 The idea: two pizza outlets

The idea: two pizza outlets



Round 1 Round 2 Round 3: stop

Summary

Order from the nearest outlet. **Move** each outlet to the average of its customers.

Slide 6: order from the nearest outlet, then move each outlet to the average of its customers. Three rounds and it stops.

Slide 7 Lloyd's algorithm

Lloyd's algorithm

Algorithm 1

Input: points $\mathbf{x}_1, \ldots, \mathbf{x}_n$; K; initial centroids

1: **repeat**
2: $\quad g_i \leftarrow \arg\min_k \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2$ for every i — assign
3: $\quad \boldsymbol{\mu}_k \leftarrow$ mean of cluster k, for every k — update
4: **until** no label changes



Start (deliberately bad): $\mu_A = (2, 3)$, $\mu_B = (1, 3)$.

Slide 7: the algorithm, and a deliberately poor start at $(2, 3)$ and $(1, 3)$.

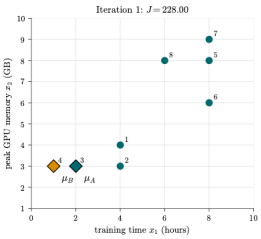
Slide 8 Iteration 1: assign

Iteration 1: assign

run	$\mathbf{x}_i$	$\|\mathbf{x}_i - \mu_A\|^2$ $\mu_A = (2.000, 3.000)$	$\|\mathbf{x}_i - \mu_B\|^2$ $\mu_B = (1.000, 3.000)$	assigned to
1	(4, 4)	**5.00**	10.00	A
2	(4, 3)	**4.00**	9.00	A
3	(2, 3)	**0.00**	1.00	A
4	(1, 3)	1.00	**0.00**	B
5	(8, 8)	**61.00**	74.00	A
6	(8, 6)	**45.00**	58.00	A
7	(8, 9)	**72.00**	85.00	A
8	(6, 8)	**41.00**	50.00	A

Run 1: 5 to μ_A, 10 to $\mu_B \Rightarrow$ A

$J = 228.00$



Slide 8: run 1 is at squared distance 5 from μ_A and 10 from μ_B, so it joins A . Only run 4 joins B. $J = 228$.

Slide 9 Iteration 1: update

Iteration 1: update

Cluster A = runs 1, 2, 3, 5, 6, 7, 8:

$$\mu_A = \left(\frac{40}{7}, \frac{41}{7}\right) = (5.714, 5.857)$$

Cluster B = run 4 alone: $\mu_B = (1, 3)$.

Insight

μ_A is dragged towards the large runs.

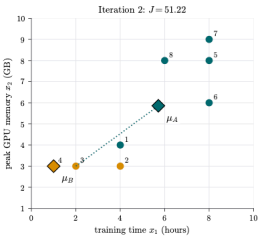
Slide 9: μ_A moves to $(5.714, 5.857)$. It is dragged towards the large runs, since most of its points are large.

Slide 10 Iteration 2: runs 2 and 3 move to B

Iteration 2: runs 2 and 3 move to B

run	$\mathbf{x}_i$	$\|\mathbf{x}_i - \mu_A\|^2$ $\mu_A = (5.714, 5.857)$	$\|\mathbf{x}_i - \mu_B\|^2$ $\mu_B = (1.000, 3.000)$	assigned to
1	(4, 4)	**6.39**	10.00	A
2	(4, 3)	11.10	**9.00**	B
3	(2, 3)	21.96	**1.00**	B
4	(1, 3)	30.39	**0.00**	B
5	(8, 8)	**9.82**	74.00	A
6	(8, 6)	**5.24**	58.00	A
7	(8, 9)	**15.10**	85.00	A
8	(6, 8)	**4.67**	50.00	A

J: 228.00 → 51.22

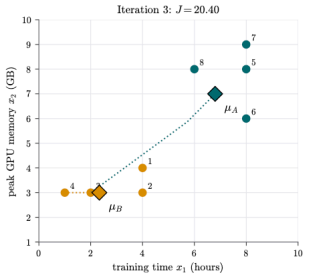


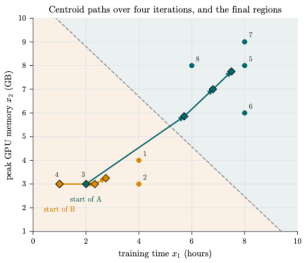
Slide 10: runs 2 and 3 switch sides, and J falls from 228 to 51.22.

Slide 11

Iterations 3 and 4: stop

Iterations 3 and 4: stop





J: 228.00 → 51.22 → 20.40 → 15.25 Best of all 256.

Slide 11: J falls $228 \to 51.22 \to 20.40 \to 15.25$, which is the best of all 256 labellings.

Slide 12

Update step: the best centroid is the mean

Update step: the best centroid is the mean

One cluster, one coordinate, values $x_1, \ldots, x_m$:

$$f(c) = \sum_{j=1}^{m} (x_j - c)^2$$

$$f'(c) = -2 \sum_{j=1}^{m} (x_j - c) = 0 \implies \boxed{c = \frac{1}{m} \sum_{j=1}^{m} x_j}$$

$f''(c) = 2m > 0$: a minimum.

Small runs, first coordinate $4, 4, 2, 1$ (mean 2.75):
$f(2) = 9.00, \quad f(2.75) = 6.75, \quad f(3.5) = 9.00.$

Slide 12: the derivative gives the mean, and the second derivative $2m > 0$ confirms a minimum. See A4.

Slide 13 Why squared distance?

Why squared distance?

- Squared $\Rightarrow$ mean (cheap)
- Absolute $\Rightarrow$ median (K-medians)

Warning

Outliers pull the mean: $1, 2, 3, 100$ has mean 26.5, median 2.5.

Slide 13: squared distance gives the mean, absolute distance gives the median. Outliers pull the mean.

Slide 14 Assignment step: the boundary is a straight line

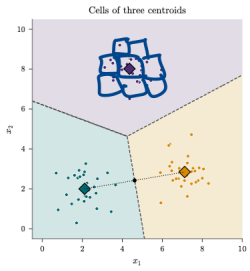
Assignment step: the boundary is a straight line

$\mathbf{x}$ is nearer $\boldsymbol{\mu}_a$ than $\boldsymbol{\mu}_b$ when

$$2(\boldsymbol{\mu}_b - \boldsymbol{\mu}_a)^\top \mathbf{x} < \|\boldsymbol{\mu}_b\|^2 - \|\boldsymbol{\mu}_a\|^2$$

Linear in $\mathbf{x}$: the perpendicular bisector.

Eight runs: $9.5\,x_1 + 9\,x_2 = 98.1875$



Slide 14: the boundary is the perpendicular bisector, here $9.5x_1 + 9x_2 = 98.1875$. See A8.

Slide 15 J never increases, so the algorithm stops

J never increases, so the algorithm stops

- **Assign:** nearest centroid $\Rightarrow$ J cannot rise
- **Update:** mean is best $\Rightarrow$ J cannot rise
- Finitely many labellings $\Rightarrow$ it stops

Warning

Stopping does not mean best.
Numbers $0, 1, 2, 4$ with $K = 2$:
it can stop at $\{0,1\}$ $\{2,4\}$: $J = 0.25 + 0.25 + 1 + 1 = 2.5$
yet better is $\{0,1,2\}$ $\{4\}$: $J = 1 + 0 + 1 + 0 = 2$

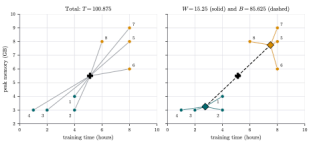
Slide 15: neither step raises J, but stopping is not the same as being best. See A10.

Slide 16 Within-cluster and between-cluster spread

Within-cluster and between-cluster spread: $T = W + B$

$$\underbrace{\sum_i \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2}_{T} = \underbrace{\sum_i \|\mathbf{x}_i - \boldsymbol{\mu}_{g_i}\|^2}_{W=J} + \underbrace{\sum_k n_k \|\boldsymbol{\mu}_k - \bar{\mathbf{x}}\|^2}_{B}$$

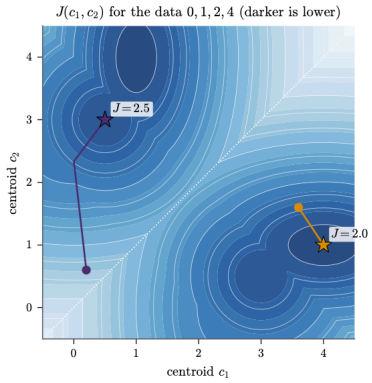
- W: spread inside clusters
- B: spread between cluster means
- T fixed $\Rightarrow$ smaller W = larger B
- Eight runs:
  $15.25 + 85.625 = 100.875$



Slide 16: $T = W + B$, with $100.875 = 15.25 + 85.625$. See A9.

Slide 17 J has several valleys

J has several valleys



Data $0, 1, 2, 4$, $K = 2$.

Stars: the two stopping points, $J = 2.5$ and $J = 2$.

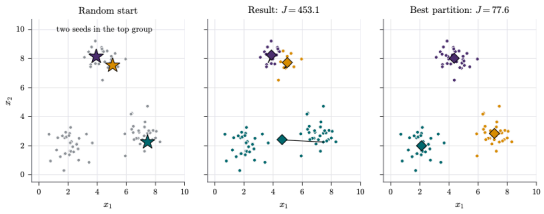
Local minimum: no step lowers J, yet a better answer exists.

Best answer: NP-hard to find.

Slide 17: stars mark the two stopping points for $0, 1, 2, 4$, with $J = 2.5$ and $J = 2$.

Slide 18 A random start can fail

A random start can fail



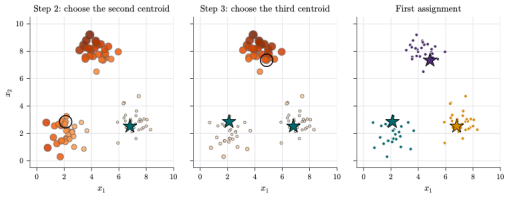
- Two centroids in one group: 77% of random starts
- Result $J = 453.1$ vs best 77.6

Slide 18: two centroids in one group happen in 77% of random starts, giving $J = 453.1$ against the best 77.6.

Slide 19

The fix: spread the starting centroids out

The fix: spread the starting centroids out



1. First centroid: a random data point (star).
2. D_i = squared distance from point i to its nearest chosen centroid.
3. Next centroid: draw a point with probability $\propto D_i$ (bigger disc = likelier). Repeat.

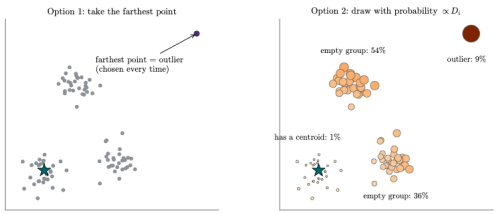
Slide 19: bigger discs mark points more likely to be chosen next.

Slide 20

Why not simply take the farthest point?

Why not simply take the farthest point?

Goal: next centroid in an empty group. Star = first centroid.



- Farthest point: always the outlier $\Rightarrow$ centroid wasted.
- Drawing: 30 far points outweigh 1 $\Rightarrow$ 90% in an empty group.

Slide 20: the farthest point is always the outlier, while drawing at random picks the empty group 90% of the time.

Slide 21 K-means++

K-means++

Algorithm 2: K-means++ initialisation

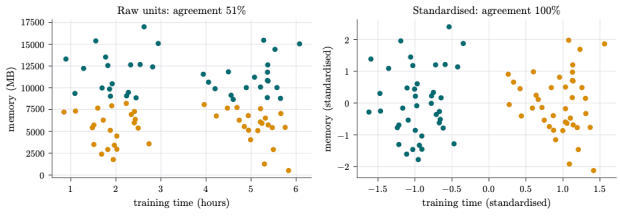
1: $\boldsymbol{\mu}_1 \leftarrow$ a data point chosen at random
2: **for** $k = 2, \ldots, K$ **do**
3: $\quad D_i \leftarrow$ squared distance from $\mathbf{x}_i$ to its nearest chosen centroid
4: $\quad \boldsymbol{\mu}_k \leftarrow \mathbf{x}_i$ with probability $D_i / \sum_l D_l$
5: **end for**
6: run Lloyd's algorithm from $\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_K$

90 points, 400 starts each: best partition reached random 84% vs K-means++ 97.8%

Slide 21: over 400 starts on 90 points, random starts reach the best partition 84% of the time and K-means++ 97.8%.

Slide 22 Standardise the features first

Standardise the features first



Hours vs megabytes: agreement 51% → 100% after standardising

Slide 22: agreement rises from 51% to 100%.

Slide 23 How many clusters, K?

How many clusters, K?

K = number of clusters. We choose it before running K-means.

Smallest J for the eight runs:

K	1	2	3	4	5	6	7	8
best J	100.875	15.250	8.750	4.333	1.500	1.000	0.500	0.000

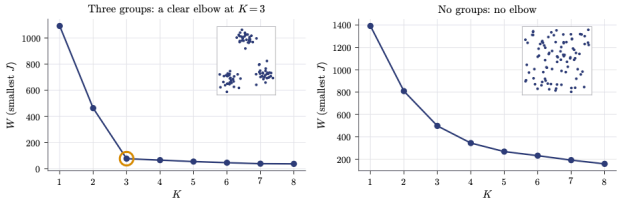
More clusters $\Rightarrow$ smaller J, down to $J = 0$ at $K = n$. So J alone cannot choose K.

Slide 23: more clusters always means a smaller J, so J alone cannot choose K.

Slide 24 1. Elbow method

1. Elbow method

W_K = smallest J with K clusters. **Elbow:** the K after which W_K stops falling fast.



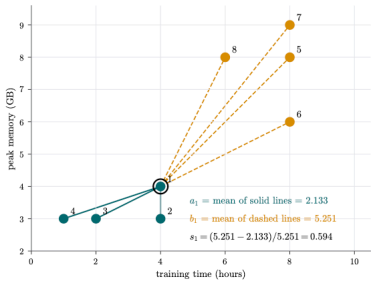
Drops: 628.1, 386.3, 11.2 $\Rightarrow$ bend at $K = 3$. Weakness: often no clear bend (right).

Slide 24: drops of 628.1, 386.3 and then 11.2 give a bend at $K = 3$. The right panel shows the weakness: often no clear bend.

Slide 25

2. Silhouette of one point

2. Silhouette of one point



a_i = mean distance to the other points of its own cluster

b_i = mean distance to the points of the nearest other cluster

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

$s_i \approx 1$: well placed
$s_i \approx 0$: on a border
$s_i < 0$: probably in the wrong cluster

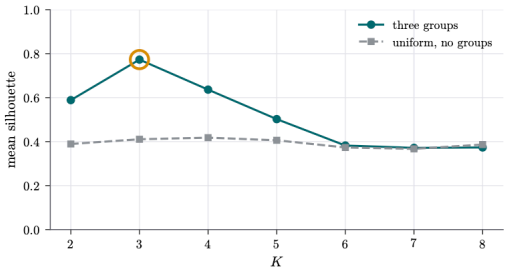
Slide 25: a_i looks inward, b_i looks at the nearest other cluster.

Slide 26

Choosing K with the silhouette

Choosing K with the silhouette

Silhouette score = mean of s_i over all points. Pick the K with the largest score.



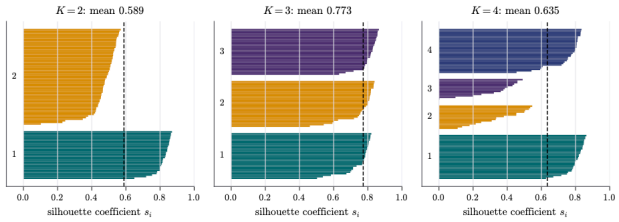
Slide 26: pick the K with the largest mean silhouette.

Slide 27

Silhouette plot: which cluster is wrong

Silhouette plot: which cluster is wrong

One bar per point, sorted within each cluster. Dashed line: silhouette score.



Slide 27: one bar per point, sorted within each cluster. It shows which cluster is responsible for a low score.

Slide 28

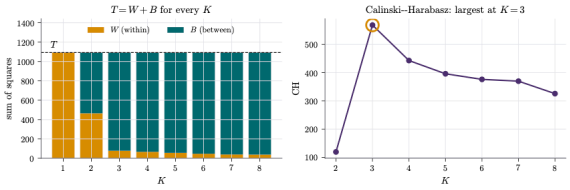
3. Calinski-Harabasz

3. Calinski–Harabasz

$$\mathrm{CH}(K) = \frac{B/(K-1)}{W/(n-K)}$$

B: spread between cluster means
W: spread inside clusters
$(T = W + B)$

$K = 3$: $507.2/0.892 = 568.8$
$K = 4$: $341.9/0.772 = 443.1$
Pick the largest: $K = 3$.

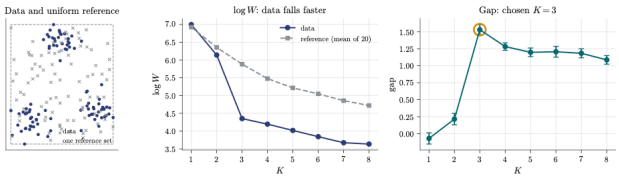


Slide 28: $K = 3$ gives $507.2/0.892 = 568.8$ and $K = 4$ gives $341.9/0.772 = 443.1$, so pick $K = 3$.

Slide 29

4. Gap statistic

4. Gap statistic



$$\mathrm{Gap}(K) = \overline{\log W_K^*} - \log W_K$$

W_K^*: the same W_K computed on random uniform data in the same box (20 copies).
Pick the K where the data beat random data by the most: here $K = 3$.

Slide 29: compare against 20 copies of uniform data in the same box.

Slide 30

5. Domain knowledge

5. Domain knowledge

- 3 marketing budgets $\Rightarrow K = 3$
- 4 T-shirt sizes $\Rightarrow K = 4$
- 2 machine types $\Rightarrow K = 2$
- 16-colour image $\Rightarrow K = 16$

Insight

Often the application decides K.

Slide 30: 3 budgets, 4 T-shirt sizes, 2 machine types, a 16-colour image.

Slide 31 Limitations of K-means

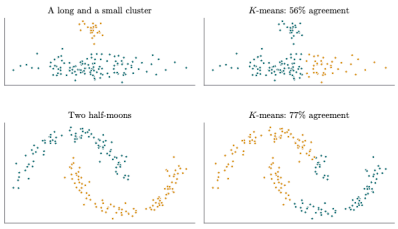
Limitations of *K*-means

Limitation	Where it bites
K must be chosen	data with no clear groups
Result depends on the start	a random start split a group
Depends on feature units	hours vs megabytes
Outliers pull centroids	one extreme run drags a mean
Round, similar clusters; straight boundaries	long clusters, half-moons
Numbers and Euclidean distance only	categories, e.g. GPU type

Slide 31: see the table in A14.

Slide 32 Where K-means fails: shape

Where *K*-means fails: shape



Long + small cluster: 56% agreement

Half-moons: 77% agreement

Boundaries are straight lines.

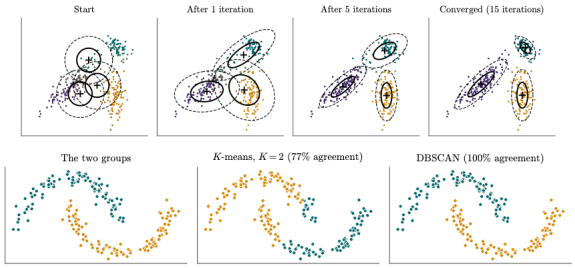
Slide 32: a long cluster with a small one gives 56% agreement, and half-moons give 77%.

Slide 33 What handles these cases

What handles these cases

Gaussian mixtures
each cluster an ellipse

DBSCAN
clusters as dense chains



Slide 33: Gaussian mixtures make each cluster an ellipse, and DBSCAN makes clusters out of dense chains. These are the next note.

Slide 34 Live demonstration

Live demonstration

`unsupervised_demo.ipynb`, Part A

1. Pizza outlets, round by round
2. Random vs K-means++ starts
3. Elbow and silhouette
4. Round groups, then half-moons

Slide 34: the notebook covers pizza outlets, random and K-means++ starts, elbow and silhouette, and round groups then half-moons.

Slide 35

Review questions

Review questions

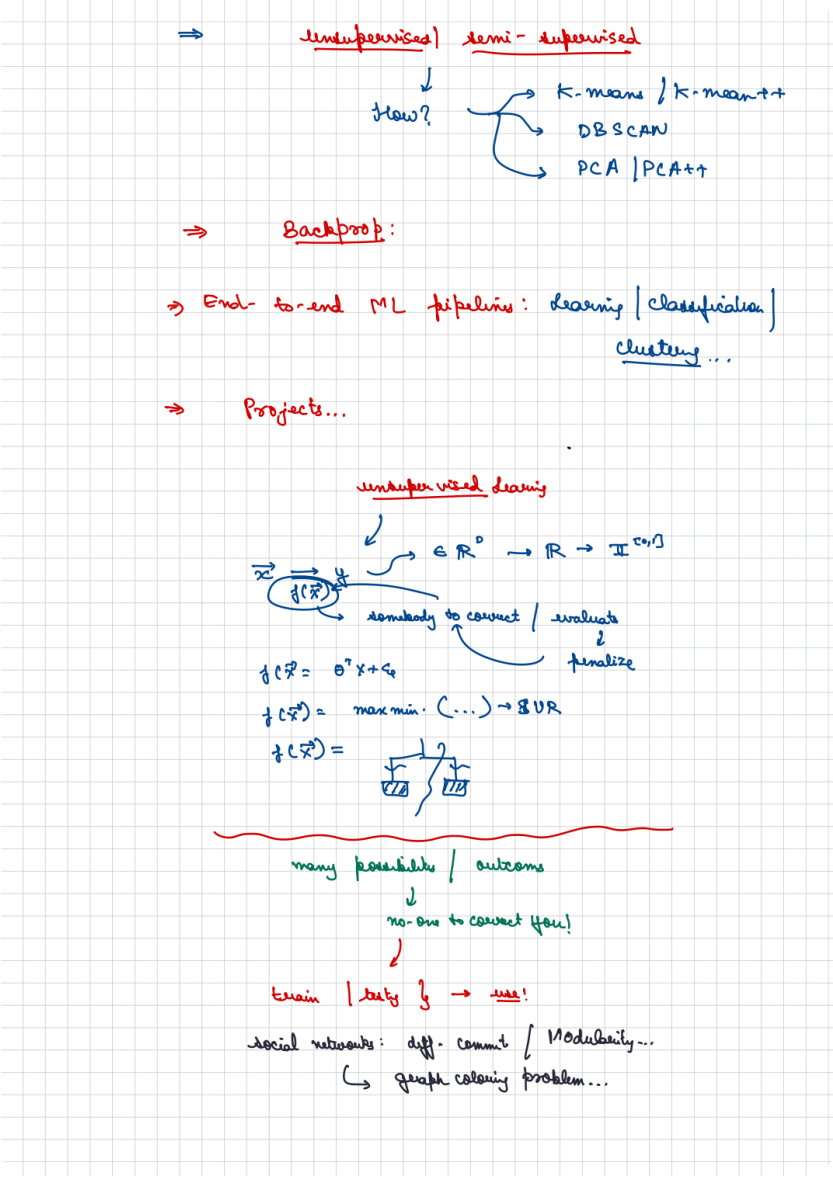
1. Why can the update step not increase J?
2. Points $1, 2, 9$, start at 1 and 2: final J?
3. Why does the smallest J never rise with K?
4. Why random, not farthest, in K-means++?
5. Why divide by $K - 1$ and $n - K$ in CH?

Handbook Chapter 2, Problems 2.1–2.15.

Slide 35: the review questions are answered in Part C.

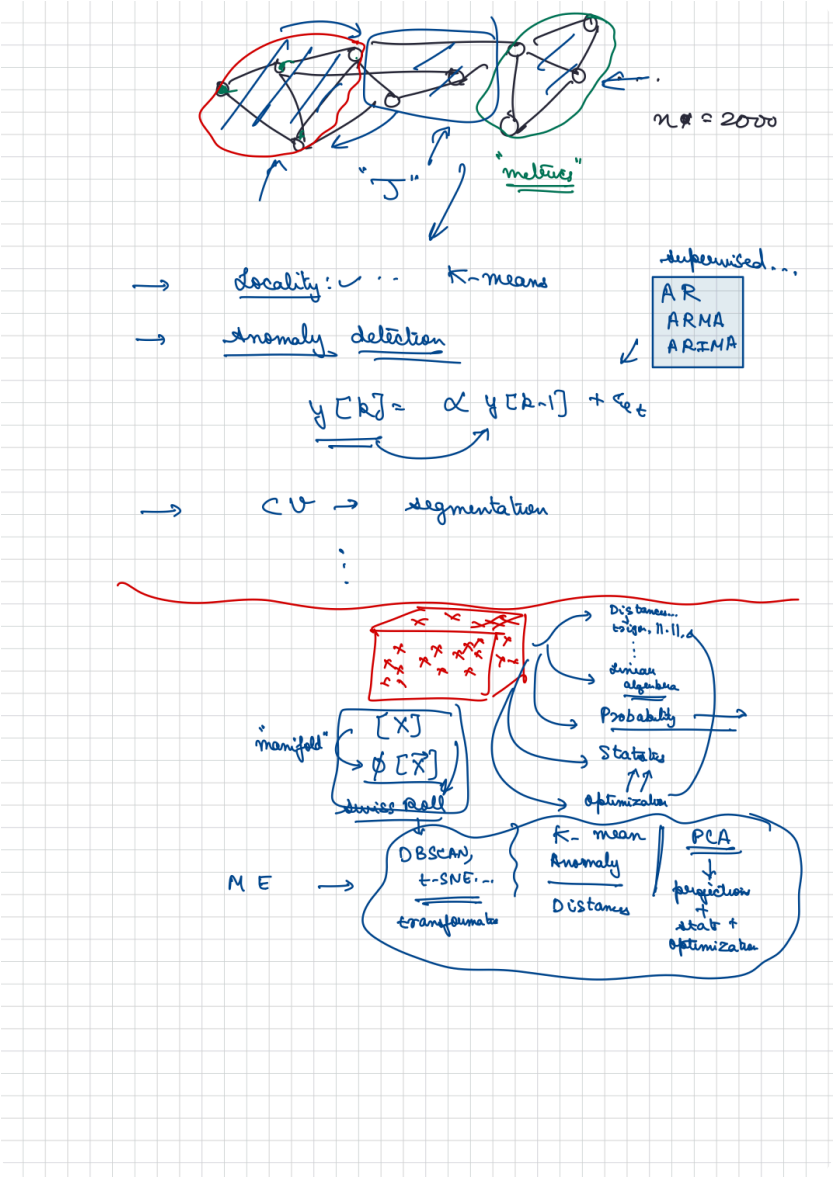
Board notes — What was written in class

Board p. 1 — Why unsupervised learning



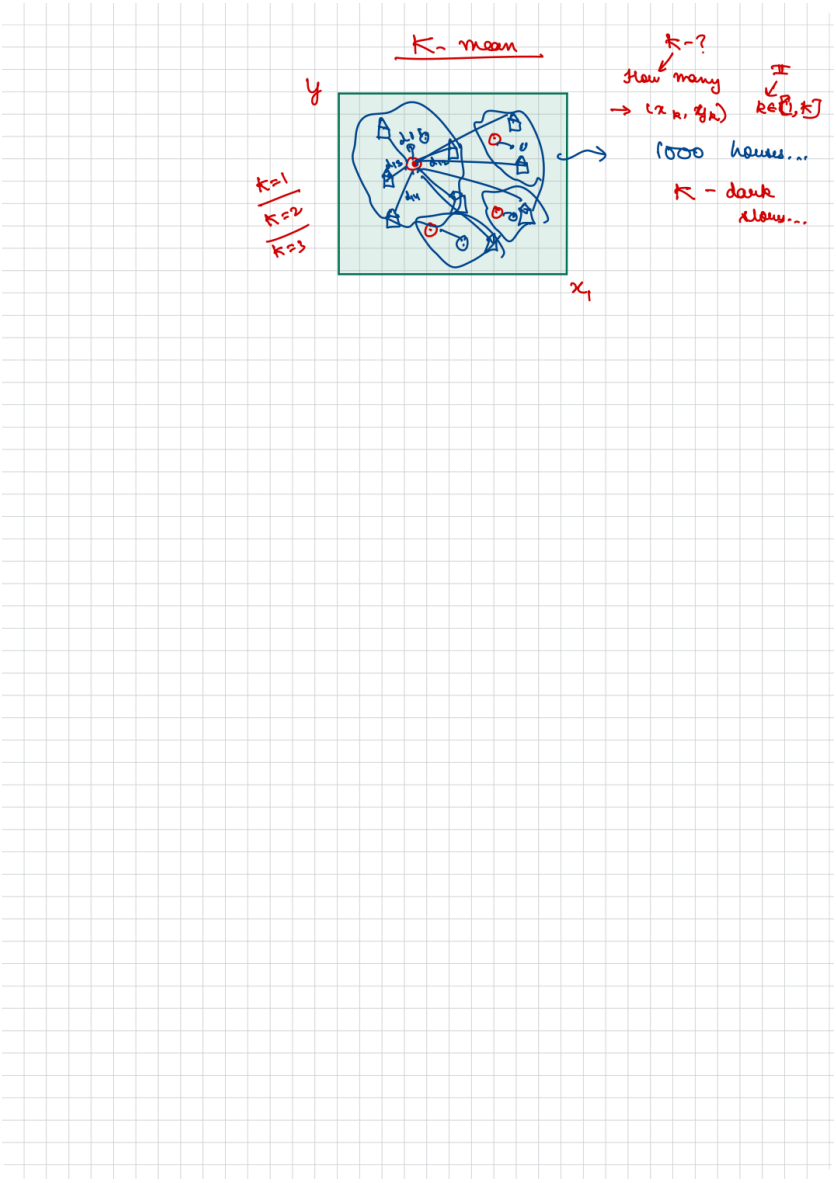
In supervised learning somebody can correct or penalise the model. In unsupervised learning there are many possible outcomes and no one to correct you. Social networks are given as an example: communities, modularity and graph colouring.

Board p. 2 — Where this shows up



Communities in a graph of about 2000 nodes, locality, anomaly detection using a time-series model such as $y[k] = \alpha y[k-1] + \epsilon_k$ (AR, ARMA, ARIMA), image segmentation, and data on a manifold such as the Swiss roll. Methods split into DBSCAN and t-SNE on one side, and K-means, anomaly detection and PCA on the other.

Board p. 3 — K-means as houses



A thousand houses on a map and a choice of K dark stores. Each house is served by its nearest store, and the question is how many stores, and where.

Part C Practice

P1. Data $1, 2, 9$ with $K = 2$, starting at centroids 1 and 2. Find the final clusters and J.

Assign: $\{1\}$ and $\{2, 9\}$, so the centroids become 1 and 5.5. Assign: 2 is at distance 1 from 1 and 3.5 from 5.5, so it moves. Centroids 1.5 and 9. Nothing changes. $J = 0.25 + 0.25 + 0 = 0.5$.

P2. Why can the update step not increase J?

With the labels fixed, J separates over clusters and coordinates, and the mean is the exact minimiser of each piece.

P3. Why does the smallest J never rise with K?

Take the best partition into K clusters and move one point into its own new cluster. That point's term becomes zero, and removing a point from a cluster and recomputing its mean cannot raise that cluster's sum.

P4. Two runs differ by 3 hours and 2000 MB. Standard deviations are 1.6 hours and 3634 MB. Find memory's share of the squared distance, raw and standardised.

Raw: $9 + 4{,}000{,}000$, so memory's share is about 99.9998%. Standardised: $(3/1.6)^2 = 3.516$ and $(2000/3634)^2 = 0.303$, so its share is $0.303/3.819 = 7.9\%$.

P5. K-means++ on $0, 1, 2, 10, 11$ with first centroid 0. Give the probabilities for the second centroid. If 10 is chosen, give those for the third.

$D = 0, 1, 4, 100, 121$, summing to 226, so $1/226$, $4/226$, $100/226$ and $121/226$. With centroids 0 and 10: $D = 0, 1, 4, 0, 1$, summing to 6, so 1 has $1/6$, 2 has $4/6$ and 11 has $1/6$.

P6. Data $2, 4, 6, 10, 12$ with $K = 2$. Find J for $\{2, 4, 6\}, \{10, 12\}$ and for $\{2, 4\}, \{6, 10, 12\}$. Is the second a fixed point?

First: $8 + 2 = 10$. Second: $2 + 18.67 = 20.67$. It is not a fixed point, because the centroids are 3 and 9.33, and 6 is closer to 3.

P7. Find all fixed points of Lloyd's algorithm for $0, 1, 2, 4$ with $K = 2$.

$\{0, 1\}, \{2, 4\}$ with $J = 2.5$, and $\{0, 1, 2\}, \{4\}$ with $J = 2$. The split $\{0\}, \{1, 2, 4\}$ is not fixed: its centroids are 0 and 2.33, and 1 is nearer to 0.

P8. Why divide by $K - 1$ and $n - K$ in Calinski-Harabasz?

B tends to grow and W tends to shrink as K rises, whether or not structure exists. Dividing by the degrees of freedom removes that automatic trend.